## 9.8.2 Area of the region bounded by a curve, y– axis and the lines $y = c$ and $y = d$.

### Case (iv)

Let $x = f(y),\ c \le y \le d$ be the equation of the portion of the continuous curve that lies to the right side of $y-$axis (that is, the portion lies either in the first quadrant or in the fourth quadrant). Then, $x \ge 0$ for every point of the portion of the curve. It is important to note that $x$ does not change its sign in the region.

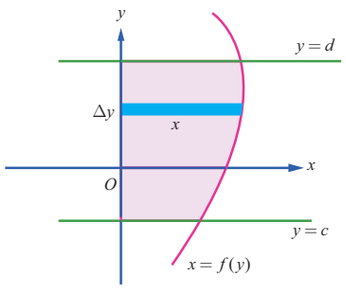


Fig. 9.11

Consider the region bounded by the curve, $y-$axis, the lines $y = c$ and $y = d$. The region is sketched as in Fig. 9.11. Then, the area $A$ of the region is found as follows:

Viewing in the positive direction of the $x-$axis, divide the region into thin horizontal strips (thin rectangles) of length $x$ and width $\Delta y$. Then, $A$ is the limit of the sum of the areas of the horizontal strips. Hence, we get $A = \lim \sum_{c \le y \le d} x\Delta y = \int_c^d x\,dy$.

### Case (v)

Let $x = f(y),\ c \le y \le d$ be the equation of the portion of the continuous curve that lies to the left side of $y-$axis (that is, the portion lies either in the second quadrant or in the third quadrant). Then, $x \le 0$ for every point of the portion of the curve. It is important to note that $x$ does not change its sign in the region. Consider the region bounded by the curve, $y-$axis, the lines $y = c$ and $y = d$. The region is sketched as in Fig. 9.12. Then, the area $A$ of the region is found as follows:

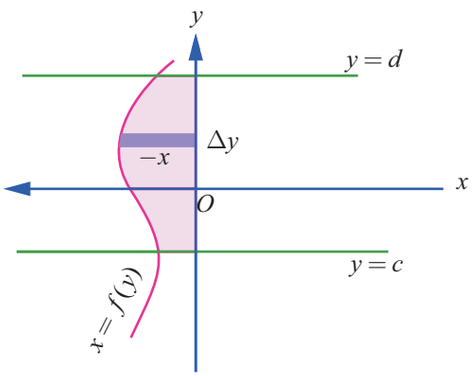


Fig. 9.12

Viewing in the positive direction of the $x-$axis, divide the region into thin horizontal strips (thin rectangles) of length $|x| = -x$ and width $\Delta y$. Then, $A$ is the limit of the sum of the areas of the horizontal strips.

Hence, we get $A = \lim \sum_{c \le y \le d} (-x)\Delta y = -\int_c^d x\,dy = \left|\int_c^d x\,dy\right|$.

### Case (vi)

Let $x = f(y),\ c \le y \le d$ be the equation of the portion of the continuous curve that lies to the right as well as to the left of the $y-$axis (that is, the portion may lie in all quadrants). Draw the graph of $x = f(y)$ in the $XY-$ plane. The graph lies alternately to the right and to the left of the $y-$axis and it is intercepted between the lines $y = c$ and $y = d$. Divide the interval $[c, d]$ into subintervals $[c, a_1],\ [a_1, a_2], \cdots,\ [a_k, d]$ such that $f(y)$ has the same sign on each of subintervals. Applying cases



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